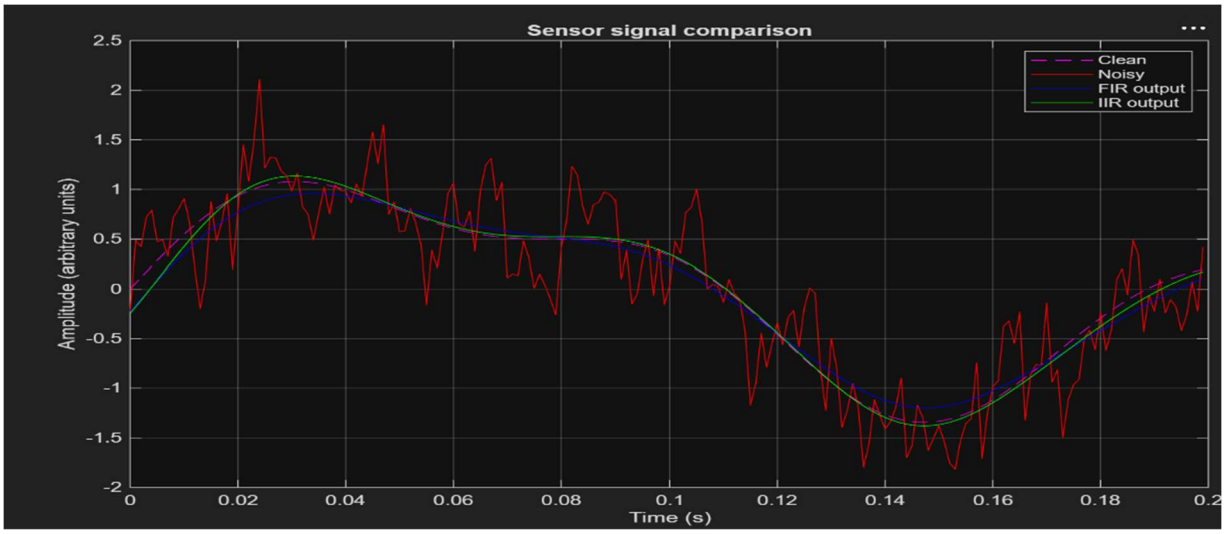
Practical Digital Filtering

I compered the FIR and IIR low-pass filter on a signal and reduced a 50 Hz hum.

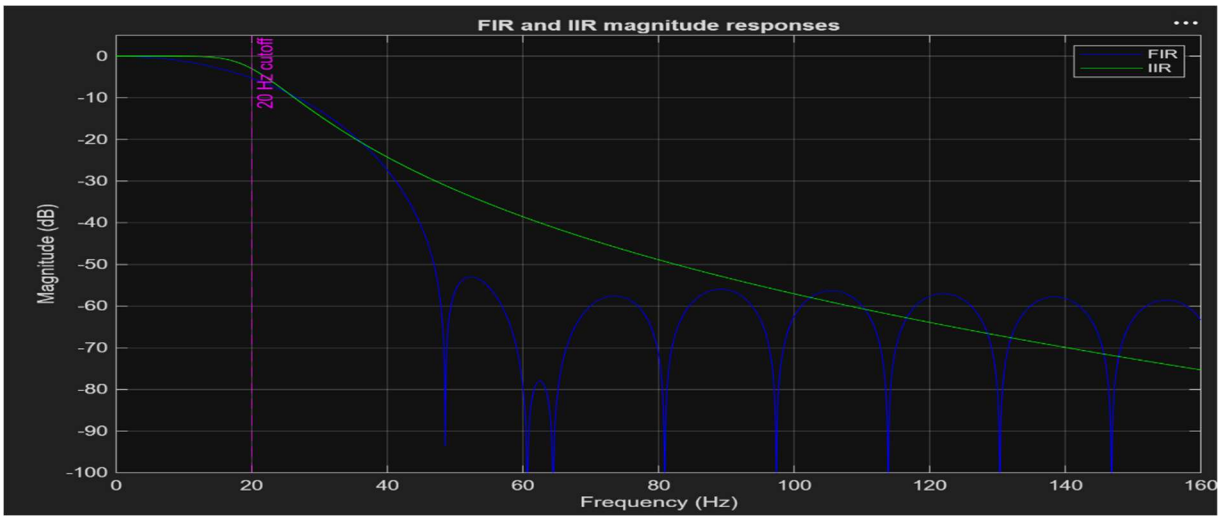
Part A: Filter a sensor signal

The signal which is two seconds long was sampled with 1000 Hz. The Signals reference contained 5 Hz and 12 hz sinus. The Amplitudes are 1 and 0.35. I added 50 Hz hum and a 137 Hz interference.

Clean, noisy and both filtered outputs over the first 0.2s:



Single-pass responses, cutoff 20 Hz:



Amplitude spectra of noisy input and filtered sensor signals:

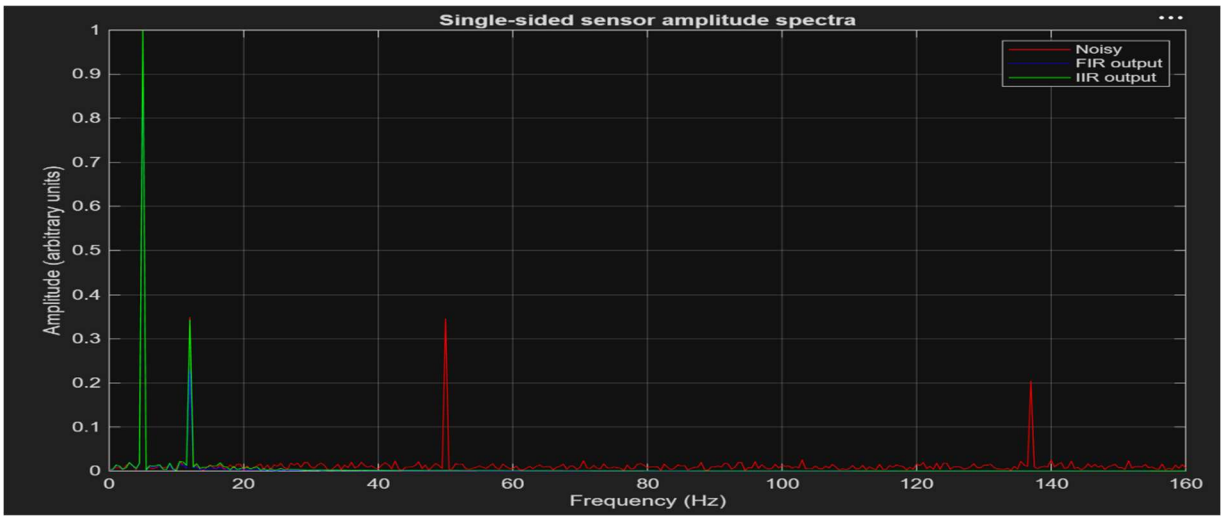


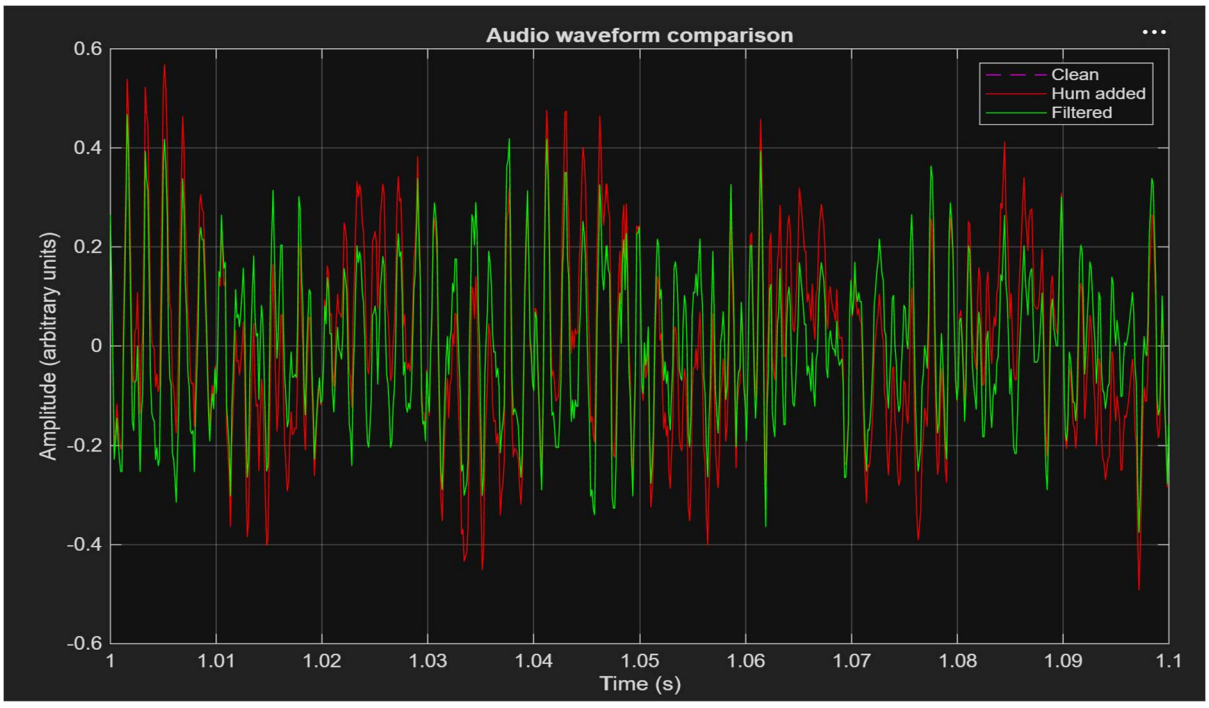
Table 1: RMSE

Signal	RMSE
Noisy input	0.38012
FIR output	0.10559
IIR output	0.054774

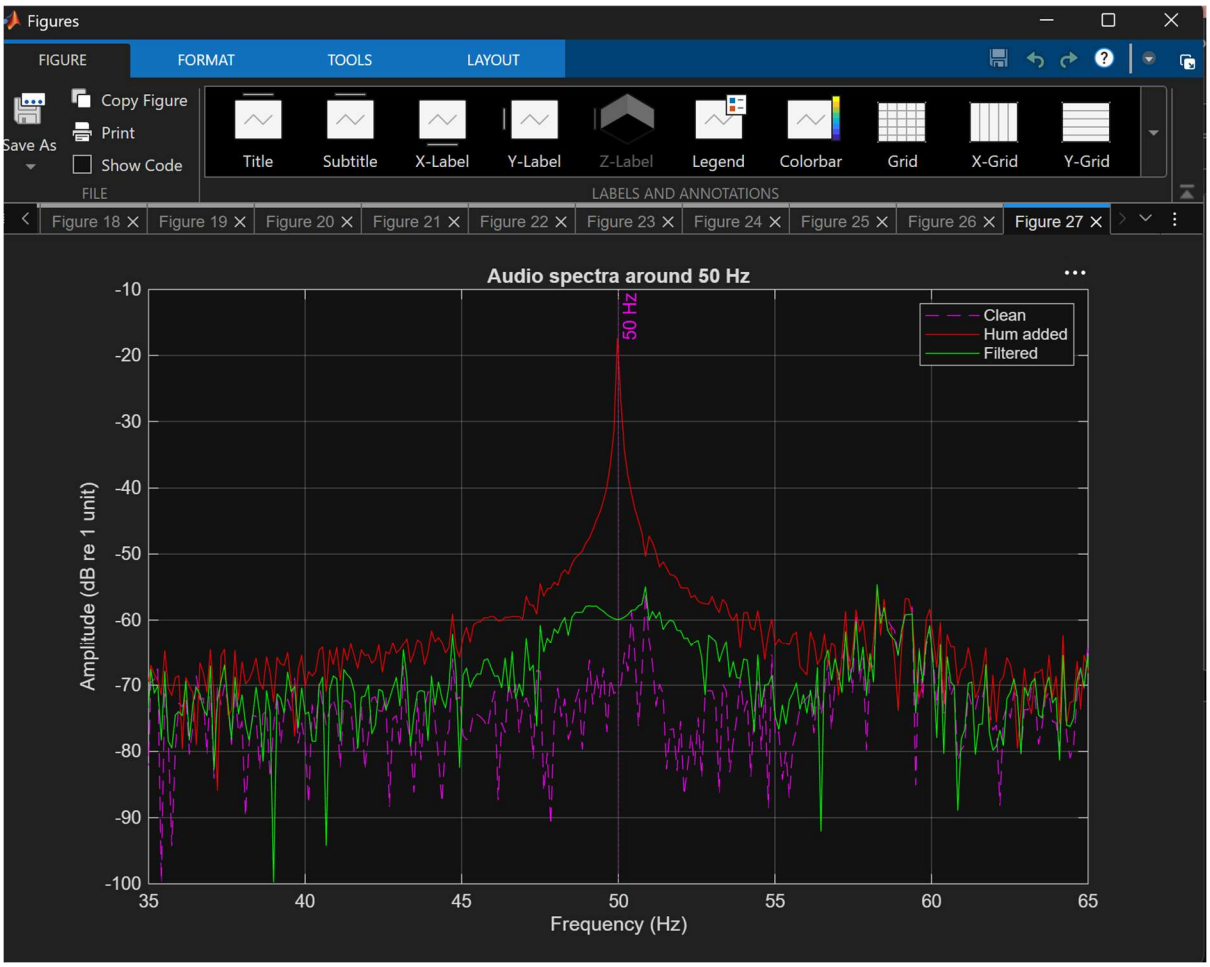
RSME We can see that both filters the overall error reduced. Further we can see that the IIR filter the least overall error has.

Part B Reduce a 50 Hz hum in audio

Clean, Hum added and filtered audio:



Single sided audio spectra (50 Hz):



Recording or stop band	50 Hz amplitude	Attenuation
Hum added	0.150195	
49-51 Hz filter	0.001027	43.30 dB
45-55 Hz filter	0.000245	55.75 dB

Questions

1. Why is a 20 Hz low-pass reasonable for Part A, and what assumption does it make about the wanted sensor signal?

Using a 20 Hz low-pass filter makes sense here. At 5 and 12 Hz we have the signals we want to keep, while the interference is at 50 and 137 Hz. With this filter, the higher frequencies gets weaker and the wanted signals can still pass. But above 20 Hz there should be no important sensor information, otherwise we lose that one too. How good it keeps the wanted signals depends also on the transition of the filter.

2. Which filter gives the better result for the requirements you chose? Support the answer with at least two kinds of evidence.

The Butterworth IIR worked better here. It keeps more of the signal that we wanted and reduces the interference. Its RMSE was 0.054774, for the FIR it was 0.10559. We also we can see in the amplitude spectrum that the 12 Hz part stayed closer to 0.35. With the FIR filter, this decreased to around 0.23. The interference peaks was reduced a lot by both filters.

3. How would your implementation change if the sensor had to be filtered live with low latency?

I probably would use filter(b,a,x) instead of filtfilt, because then each output uses only the current and past samples. As explained in the lecture, filtfilt needs the complete recording for processing forward and backward. So it cannot give a immediate live output. For low latency I would consider a low-order IIR. But we still need to check its delay, how it changes the phase, how much calculations it needs and its stability.

4. Why is a narrow band-stop appropriate for a steady 50 Hz hum? What would you do if the interference frequency drifted?

When the interference stays at 50 Hz there is a narrow band-stop from 49 to 51 Hz makes sense to reduce the hum. The useful audio frequencies that are more close to it are geting less affected this way. If the interference frequency starts to move, I would look at the spectrum and adjust the stop band to the measured interference. For example making it wider to 45–55 Hz, like it was tested in the script. With a wider band we can handle bigger changes of the hum, but some useful sound at low frequencies could get removed too.

5. Describe one limitation of your experiments and one improvement you would make in a real measurement system.

One problem is that we used artificial interference at fixed frequencies and also had a known clean signal for comparison. For real measurements I would first look at the actual spectrum. Then choose the filter limits based on the useful signal and the measured interference. With plots before and after filtering, we can check how good the useful parts are kept. This follows the steps from the lecture, where we use measurements to decide how to design the filter.